

Introduction to Machine Learning with Scikit-Learn

Learn how computers discover patterns, make predictions, and solve real-world problems using Python.

1. What is Machine Learning?

Imagine sorting delicious Indian snacks like **Jalebi** and **Dosa**. Both look different, taste different, and have unique shapes! Humans instantly recognize which is which because we have seen many examples over time.

Machine Learning (ML) is the science of teaching computers to learn directly from examples and data, rather than following rigid, predefined rules. Just like students learn by practicing problems, machines discover underlying patterns on their own!

2. Traditional Programming vs. Machine Learning

To understand Machine Learning, let's compare it with traditional computer programming:

Traditional Programming

How it works: Programmers write strict, explicit rules and instructions for the computer to execute.

- **Example:** The rules of an IPL cricket match are fixed. A score algorithm adds runs based on set rules.
- **Limitation:** The computer cannot adapt if something new happens that wasn't coded in advance.

Machine Learning

How it works: The computer is fed historical data (examples) and discovers the rules or patterns automatically.

- **Example:** Sorting Alphonso mangoes by weight, softness, and color without writing manual rules.
- **Advantage:** The system improves and adapts as it sees more data.

3. Core Building Blocks: Features (X) & Labels (y)

When a machine looks at a dataset, it separates the information into two critical components:

Features (X) — The Clues

Features are the input variables or properties that describe the situation. They provide the clues the machine uses to learn.

Cricket Match Example: Overs remaining, Runs required, Wickets in hand.

Label (y) — The Answer

The label is the target outcome or prediction value that the machine is trying to learn and predict.

Cricket Match Example: Final Match Outcome (**Win** or **Lose**).

4. Supervised Learning & Practical Example

Supervised Learning is like learning with a teacher or a cricket coach. The teacher provides the training data containing both the clues (Features X) and the correct answers (Labels y). The machine studies these pairs to learn the relationship.

Real-World Application: Sorting Alphonso Mangoes

A fruit vendor wants to classify mangoes automatically as **Ripe** or **Raw**.

- **Features (X):** Colour Score (how yellow/orange) and Softness Level.
- **Label (y):** Ripe or Raw.

The supervised ML model learns the exact color threshold and softness combination that distinguishes a sweet ripe Alphonso from a raw one!

5. The Scikit-Learn 3-Step Recipe

In Python, the `scikit-learn` library makes training a machine learning model as easy as following a 3-step recipe:

1 Import & Instantiate Model: Choose an algorithm (e.g., `DecisionTreeClassifier`) and create the model object.

2 Train Model (Fit): Call `model.fit(X, y)` so the machine learns patterns from the features and labels.

3 Make Predictions: Pass new feature data into `model.predict(X_new)` to get predictions!

Code-Along: Predicting Cricket Team Selection

Here is the full Python script to predict whether a player will be selected for the team based on practice hours and batting average:

```
from sklearn.tree import DecisionTreeClassifier

# Step 1: Prepare Training Data (Features X: [Practice Hours, Batting Avg])
X = [[5, 30], [3, 15], [8, 45], [2, 10]]
y = ['Selected', 'Not Selected', 'Selected', 'Not Selected']

# Step 2: Initialize and Fit the Model
model = DecisionTreeClassifier()
model.fit(X, y)

# Step 3: Predict for a new player (6 practice hours, 35 batting avg)
new_player = [[6, 35]]
prediction = model.predict(new_player)

print("Prediction:", prediction[0]) # Output: Selected
```

6. Model Evaluation, Bias, and Data Quality

Machine learning models are only as good as the data used to train them. Mistakes and poor predictions happen when training data is biased, incomplete, or corrupted.

Garbage In, Garbage Out!

If we train a cricket prediction model using data only from right-handed batters, the machine may fail or make unfair decisions when evaluating left-handed players.

- **Biased Data:** Leads to unfair and skewed predictions.
- **Missing / Bad Data:** Causes inaccurate model performance.
- **Key Takeaway:** Always ensure your dataset is balanced, clean, and representative!

Take-Home Mission!

Scenario: Build a mental ML model to predict whether it will rain during cricket practice tomorrow in Mumbai! Identify what features you would collect (e.g., humidity, cloud cover, wind speed) and what label you want to predict (Rain / No Rain).

Practice Questions

Test your understanding of Machine Learning concepts, Scikit-Learn workflows, and data ethics.

Instructions: Answer the following questions in the spaces provided below. Refer back to your lesson notes if needed.

Question 1: Conceptual Understanding

Explain the key difference between Traditional Programming and Machine Learning using a real-world example other than cricket.

Question 2: Features vs. Labels

In an ML application designed to predict whether a student will pass an exam, identify at least two possible **Features (\$X\$)** and the target **Label (\$y\$)**.

Question 3: Scikit-Learn Syntax

What is the purpose of the `.fit(X, y)` method in Scikit-Learn, and how does it differ from `.predict(X_new)`?

Question 4: Data & Fairness

What does the phrase "*Garbage In, Garbage Out*" mean in Machine Learning? Give an example of how biased training data can lead to unfair decisions.

Question 5: Code Analysis

Suppose you have a dataset with features $X = \begin{bmatrix} 4 & 20 \\ 7 & 40 \end{bmatrix}$ representing practice hours and score, and labels $y = ['Retry', 'Pass']$. Write down the Python code lines required to train a `DecisionTreeClassifier` on this data.
